Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Verifies that x = 1 gives y = 3	AO1.1b	B1	$y = 2\times1^2 - 8\times1^{\frac{3}{2}} + 8\times1 + 1 = 3$
	Expresses $x\sqrt{x}$ as $x^{\frac{3}{2}}$	AO1.1b	B1	$y = 2x^2 - 8x^{\frac{3}{2}} + 8x + 1$
	Attempts to differentiate (at least one term correct)	AO1.1a	M1	$\frac{dy}{dx} = 4x - 12\sqrt{x} + 8$
	Correctly differentiates	AO1.1b	A1	
	Explains $\frac{dy}{dx} = 0$ for stationary or maximum point Must be explicitly seen	AO2.4	E1	For stationary point $4(\sqrt{x})^2 - 12\sqrt{x} + 8 = 0$ $\sqrt{x} = 1$ or $\sqrt{x} = 2$ x = 1 or x = 4
	Shows solution of $\frac{dy}{dx} = 0$ to give x = 1 (and x = 4) (May be awarded for work seen in (b)) or correct verification of x = 1	AO1.1b	B1	
	Differentiates again (May be awarded for work seen in (b))	AO1.1a	M1	$\frac{d^2y}{dx^2} = 4 - \frac{6}{\sqrt{x}}$
	Shows that x = 1 gives a negative value (in a correct second differential)	AO1.1b	A1	$x = 1$ gives $\frac{d^2y}{dx^2} = -2$
	Concludes that maximum point is at (1, 3). Constructs rigorous mathematical argument to show the required result. Failure to score E1 does not rule out award of this mark	AO2.1	R1	Negative so maximum when x = 1 Maximum at (1, 3)
(b)	States coordinates	AO1.1b	B1	(4, 1)
	States minimum point	AO1.1b	B1	Minimum
				Total 11 marks

Notes:

A candidate who does not handle the $x\sqrt{x}$ term can score B1 B0 M1 A0 E1 B0 M1 A0 R0

Q2.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$\frac{dy}{dx} = -\frac{2}{x^3}$
			Total 1 mark

Q3.

Marking Instructions	AO	Marks	Typical Solution
Selects appropriate technique to differentiate	AO3.1a	M1	$2(x+y-2)\left(1+\frac{dy}{dx}\right) = e^y \frac{dy}{dx}$
Differentiates term involving e^y correctly	AO1.1b	B1	$\frac{dy}{dx} = 0 \Rightarrow x+y-2=0$
Differentiates fully correctly	AO1.1b	A1	$\Rightarrow 0 = e^y - 1$
Uses $\frac{dy}{dx} = 0$	AO1.1a	M1	$y = 0$ $x = 2$
Eliminates x or y from the equation of the curve	AO1.1a	M1	
Obtains correct y CAO	AO1.1b	A1	
Obtains correct x CAO	AO1.1b	A1	
			Total 7 marks

Q4.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	7
			Total 1 mark

Q5.

Marking Instructions	AO	Marks	Typical Solution

Obtains $\frac{dy}{dx}$ for both the given curves – at least one term must be correct for each curve	AO3.1a	M1	$\frac{dy}{dx} = 6x^2 + 12x - 12$ $\frac{dy}{dx} = 60 - 12x$
States both derivatives correctly	AO1.1b	A1	
Translates problem into an inequality	AO3.1a	M1	Chris's claim is **incorrect** when
States a correct quadratic inequality FT from an incorrect $\frac{dy}{dx}$ provided both M1 marks have been awarded	AO1.1b	A1	$6x^2 + 12x - 12 \le 60 - 12x$ $2x^2 + 8x - 24 \le 0$ $x^2 + 4x - 12 \le 0$ $(x + 6)(x - 2) \le 0$ Critical values are $x = -6$ and 2 region: $x < -6$ / $-6 < x < 2$ / $x \le 2$ sign: + / − / +
Determines a solution to 'their' inequality	AO1.1a	M1	$-6 \le x \le 2$
Obtains correct range of values for x Must be correctly written with both inequality signs correct	AO1.1b	A1	Chris's claim is incorrect for values of x in the range $-6 \le x \le 2$, so he is wrong
Interprets final solution in context of the original question, must refer to Chris's claim	AO3.2a	R1	
			Total 7 marks

Q6.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Rewrites given expression with a fractional power and negative power – at least one index form must be correct	AO1.1a	M1	$y = 6x^{\frac{3}{2}} + 32x^{-1}$
	Both terms correct	AO1.1b	A1	

	Differentiates 'their' rewritten expression – at least one term correct	AO1.1a	M1	$\frac{dy}{dx} = 6 \times \frac{3}{2} \times x^{\frac{1}{2}} - 32x^{-2}$ $= 9\sqrt{x} - \frac{32}{x^2}$
	Both terms correct for 'their' expression	AO1.1b	A1F	
(b)	Finds the equation of the tangent, a clear attempt must be seen	AO3.1a	M1	When $x = 4$,
	Evaluates 'their' $\frac{dy}{dx}$ (from part **(a)**) correctly (when $x = 4$)	AO1.1b	A1F	$\frac{dy}{dx} = 9 \times 2 - \frac{32}{16} = 16$ and
	Obtains correct y value (when $x = 4$)	AO1.1b	A1	$y = 6 \times 4 \times 2 + \frac{32}{4} = 56$
	Obtains correct form of the equation of a straight line using 'their' values for y and $\frac{dy}{dx}$	AO1.1b	A1F	Tangent: $y - 56 = 16(x - 4)$ When $y = 0$,
	Deduces value required at x-axis is when y equals 0 (follow through from 'their' equation) Both coordinates needed, any form	AO2.2a	A1F	$x = 4 - \frac{56}{16} = 0.5$ $(0.5,0)$
				Total 9 marks

Q7.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Differentiates, with at least one term correct	AO1.1a	M1	$\frac{dv}{dt} = 12t - 36^2$
	Selects and applies $F = ma$ to 'their' derivative Condone use of 400 for mass	AO1.1a	M1	$F = ma = 0.4\,(12t - 36t^2)$
	Obtains correct expression for force	AO1.1b	A1F	$= 4.8t - 14.4t^2$

	FT from 'their' $F = ma$ equation, provided the first M1 has been awarded (may be in factorised form)			
(b)	Integrates v to find r, with at least one term correct	AO3.1b	M1	$r = \int (6t^2 - 12t^3)\,dt$
	Obtains correct integral (condone absence of c)	AO1.1b	A1	$r = 2t^3 - 3t^4 + c$
	Deduces the value of c using initial conditions FT use of 'their' integral provided M1 awarded	AO2.2a	A1F	When $t = 0$, $r = 0$ so $0 = 2 \times 0^2 - 3 \times 0^4 + c$ so $c = 0$
	Forms and solves an equation for t (condone numerical slip)	AO1.1a	M1	$0 = 2t^3 - 3t^4$ $= t^3(2 - 3t)$ $t = 0$ or $t = \frac{2}{3}$
	Interprets solution, realising that the non-zero time is required (Must include units) FT use of 'their' equation for t provided both M1 marks have been awarded	AO3.2a	A1F	Next at O at $\frac{2}{3}$ seconds
				Total 8 marks

Q8.

Marking Instructions	**AO**	**Marks**	**Typical Solution**
Selects an appropriate method – **either** differentiates, at least as far as: $\frac{dy}{dx} = 2x\ldots$ **or** commences completion of the square: $\left(x - \frac{5}{2}\right)^2 + \ldots$	AO1.1a	M1	$y = \left(x - \frac{5}{2}\right)^2 - \frac{25}{4} + a$ y minimised when squared bracket is 0 $\left(\frac{5}{2}, a - \frac{25}{4}\right)$ **ALT** $\frac{dy}{dx} = 2x - 5$ so $2x - 5 = 0$ for minimum $x = \frac{5}{2}$
Fully differentiates **and** sets derivative equal to zero **or** fully completes	AO1.1a	M1	

square Allow one error			$y=\left(\frac{5}{2}\right)^2-5\left(\frac{5}{2}\right)+a=a-\frac{25}{4}$
Obtains both coordinates	AO1.1b	A1	
			Total 3 marks

Q9.

Marking Instructions	**AO**	**Marks**	**Typical Solution**
Explains clearly that $\mathrm{f}(x)$ is increasing $\Leftrightarrow \mathrm{f}'(x) > 0$ (for all values of x) or Explains $\Rightarrow \mathrm{f}(x)$ is increasing $\mathrm{f}'(x) > 0$ for all values of x This may appear at any appropriate point in their argument	AO2.4	E1	For all x, $\mathrm{f}'(x) > 0 \Rightarrow \mathrm{f}(x)$ is an increasing function $\mathrm{f}(x) = x^3 - 3x^2 + 15x - 1$ $\Rightarrow \mathrm{f}'(x) = 3x^2 - 6x + 15$ $\left.\begin{array}{l}\Rightarrow \mathrm{f}'(x) = 3(x-1)^2+12 \\ \therefore \mathrm{f}'(x) \text{ has a minimum} \\ \text{value of } 12\end{array}\right\}(*)$ therefore $\mathrm{f}'(x) > 0$ for all values of x **OR** for $\mathrm{f}'(x)$, $b^2 - 4ac = -144$ $\therefore \mathrm{f}'(x) \neq 0$ for any real x, so $\mathrm{f}'(x)$ is either always positive or always negative. $\mathrm{f}'(0) = 15$ therefore $\mathrm{f}'(x) > 0$ for all values of x **OR** $\mathrm{f}''(x) = 6x - 6$, which $= 0$ when $x = 1$ so min $\mathrm{f}'(x)$ is $\mathrm{f}'(1) = 12$ therefore $\mathrm{f}'(x) > 0$ for all values of x Thus, since, $\mathrm{f}'(x) > 0$ for all values of x it is proven that $\mathrm{f}(x)$ is an increasing function.
Differentiates – at least two correct terms	AO1.1a	M1	
All terms correct	AO1.1b	A1	
Attempts a correct method which could lead to $\mathrm{f}'(x) > 0$	AO3.1a	M1	
Correctly deduces $\mathrm{f}'(x) > 0$ (for all values of x)	AO2.2a	A1	
Writes a clear statement that links the steps in the argument together, the deduction about a positive gradient for all values of x proves that the given function is increasing for all values of x	AO2.1	R1	
			Total 6 marks

Q10.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$-\dfrac{1}{x\sqrt{x}}$
			Total 1 mark

Q11.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States the correct derivative	AO1.1b	B1	$2^x\ln 2$
(b)	Selects an appropriate method for integrating, which could lead to a correct exact solution (this could be indicated by an attempt at a substitution or attempting to write the integrand in the form $f'(x)f(x)^n$)	AO3.1a	M1	Let $u = 2^x$ Then $\dfrac{du}{dx} = 2^x \ln 2$ And $\dfrac{1}{\ln 2}\dfrac{du}{dx} = 2^x$ $I = \int (3+u)^{\frac{1}{2}} \dfrac{1}{\ln 2}\dfrac{du}{dx}\,dx$ $= \dfrac{1}{\ln 2}\int (3+u)^{\frac{1}{2}}\,du$ $= \dfrac{3}{3\ln 2}(3+u)^{\frac{3}{2}} + c$ Sub limits: $\left[\dfrac{2}{3\ln 2}(3+u)^{\frac{3}{2}}\right]_1^2$ $\dfrac{3}{2} \times \dfrac{1}{\ln 2}(5\sqrt{5} - 8)$ **ALT (direct inspection)** $\int 2^x\sqrt{3+2^x}\,dx$ $= \dfrac{1}{\ln 2}\int 2^x \ln 2\sqrt{3+2^x}\,dx$ $= \dfrac{1}{\ln 2}\int 2^x \ln 2(3+2^x)^{\frac{1}{2}}\,dx$
	Correctly writes integrand in a form which can be integrated (condone missing or incorrect limits)	AO1.1b	A1	
	Integrates 'their' expression (allow one error)	AO1.1a	M1	
	Substitutes correct limits corresponding to 'their' method	AO1.1a	M1	
	Obtains correct value in an exact form	AO1.1b	A1	
	Mark awarded if they have a completely correct solution, which is clear, easy to follow and contains no slips Substitution should be	AO2.1	R1	

clearly stated in exact form and change of variable or solution by direct inspection should be achieved correctly with correct use of symbols and connecting language			$= \frac{1}{\ln 2} \times \frac{2}{3}(3+2^x)^{\frac{3}{2}}$ $\left[\frac{1}{\ln 2} \times \frac{2}{3}(3+2^x)^{\frac{3}{2}}\right]_0^1$ $\frac{3}{2} \times \frac{1}{\ln 2}(5\sqrt{5}-8)$
			Total 7 marks

Q12.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Selects an appropriate routine procedure; evidence of quotient rule or product rule	AO1.1a	M1	$\frac{dy}{dx} = \frac{2(4x^2+7)-8x(2x+3)}{(4x^2+7)^2}$
	Obtains correct derivative (no need for simplification)	AO1.1b	A1	
(ii)	States clearly that $\frac{dy}{dx} > 0 \Rightarrow y$ is increasing	AO2.4	R1	y is increasing $\Leftrightarrow \frac{dy}{dx} > 0$ $\frac{2(4x^2+7)-8x(2x+3)}{(4x^2+7)^2} > 0$ $(4x^2+7)^2 > 0$ for all x $\therefore 2(4x^2+7) - 8x(2x+3) > 0$ $8x^2 + 14 - 16x^2 - 24x > 0$ $4x^2 + 12x - 7 < 0$ **(AG)**
	Forms inequality from 'their' $\frac{dy}{dx} > 0$	AO3.1a	B1	
	Deduces numerator must be positive	AO2.2a	R1	
	Considers denominator alone and sets out clear argument to justify given inequality **AG** Only award this mark if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	
(b)	Solves the correct quadratic inequality (accept evidence of factorising, completing the square, use of formula, or	AO1.1a	M1	$(2x + 7)(2x - 1)$ $x = -\frac{7}{2}, \frac{1}{2}$

correct critical values stated)			
Obtains fully correct answer, given as an inequality or using set notation	AO1.1b	A1	$-\frac{7}{2} < x < \frac{1}{2}$ **Or** $x \in \left(-\frac{7}{2}, \frac{1}{2}\right)$ **Or** $x \in \left\{x : -\frac{7}{2} < x < \frac{1}{2}\right\}$
			Total 8 marks

Q13.

Marking Instructions	AO	Marks	Typical Solution
Finds the difference between the maximum and minimum values of y	AO3.1b	M1	$x^2 + 2xy + 2y^2 = 10$ $2x + 2y + 2x\frac{dy}{dx} + 4y\frac{dy}{dx} = 0$ Highest and lowest points occur when $\frac{dy}{dx} = 0$ $\frac{dy}{dx} = 0 \Rightarrow x = -y$ $y^2 - 2y^2 + 2y^2 = 10$ $y = \pm\sqrt{10}$ $\therefore$ Height $= \sqrt{10} - (-\sqrt{10})$ $= 2\sqrt{10} = 6.32$m
Uses implicit differentiation	AO1.1a	M1	
Differentiates correctly	AO1.1b	A1	
States stationary points occur when $\frac{dy}{dx} = 0$	AO2.4	R1	
Uses $\frac{dy}{dx} = 0$ to find x in terms of y (or vice versa)	AO1.1a	M1	
Finds $x = -y$	AO1.1b	A1	
Deduces maximum and minimum values of y FT 'their' expression provided all M1 marks have been awarded	AO2.2a	A1F	
States the height of the sculpture above the platform FT 'their' max and min values for y provided all M1 marks have been awarded	AO2.2a	A1F	
			Total 8 marks

Q14.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Identifies and clearly defines variables.	AO3.1b	B1	Let C = total cost x = length of base edges h = length of height
	Models the cost with an expression of the form $ax^2 + bxh$	AO3.3	M1	$C = 15x^2 + 32xh$
	Eliminates either variable, using volume equation, to form a model for the cost in one variable.	AO3.3	M1	$x^2h = 60$ $h = \frac{60}{4^2}$
	Obtains a correct equation to model cost in one variable	AO3.1b	A1	$C = 15x^2 + \frac{1920}{x}$ (*) Differentiating $30x - \frac{1920}{x^2} = 0$
	Uses their model to find minimum. (at least one term correctly differentiated **and** equated to zero)	AO3.4	M1	$x^3 = 64$ $x = 4$ $h = \frac{60}{4^2} = 3.75$
	Obtains correct equation	AO1.1b	A1	Height of tank is 3.75 m
	Obtains correct value for h with correct units in context Award FT from correct substitution into incorrect equation for h but only if all three M1 marks have been awarded, must have correct units.	AO3.2a	A1F	$\frac{d^2C}{dx^2} = 30 + \frac{3840}{x^3}$ $x = 4 \Rightarrow \frac{d^2C}{dx^2} > 0$ therefore minimum **ALT from (*) onwards** $C = 900h^{-1} + 32\sqrt{60}h^{\frac{1}{2}}$ $0 = -900h^{-2} + 16\sqrt{60}h^{-\frac{1}{2}}$
	Performs a correct test of 'their' solution: uses the second derivative of 'their' expression for C in terms of x or h to justify that a minimum value for h has been found OE (Second derivative > 0 or test gradient/values either side)	AO2.4	R1	Height of tank is 3.75 m $\frac{d^2C}{dh^2} = 1800h^{-3} - 8\sqrt{60}h^{-\frac{3}{2}}$ $h = 3.75 \Rightarrow \frac{d^2C}{dh^2} = 25.6 > 0$ $\therefore$ minimum

(b) (i)	Explains that two of the sides of the tank will need	AO3.5c	E1	The sides will need to overlap to be joined, so two of the side lengths will

	to have length $x \pm 0.05$ in order to join them			need to be $x + 0.05$
(ii)	Explains that the refinement is relatively small and unlikely to have a significant effect on the result.	AO3.5a	R1	The minimum cost is likely to increase slightly, but relative to the size of the tank, an extra 5cm is unlikely to make a significant difference.
				Total 10 marks

Q15.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses a correct method for finding $\frac{dy}{dx}$ evidence for this includes sight of $\frac{dy}{dt}$ or $\frac{dx}{dt}$ and chain rule OR an attempt at implicit or explicit differentiation of a correct Cartesian equation or 'their' equation from part **(b)**	AO1.1a	M1	$\frac{dx}{dt} = 3t^2$ $\frac{dy}{dt} = 2t$ $\frac{dy}{dx} = \frac{2t}{3t^2}$ When $t = -2$ $\frac{dy}{dx} = -\frac{1}{3}$ **ALT** $y = (x-2)^{\frac{2}{3}} - 1$ $\frac{dy}{dx} = \frac{2(x-2)^{-\frac{1}{3}}}{3}$ When $t = -2$, $x = -6$ $\frac{dy}{dx} = \frac{2(-6-2)^{-\frac{1}{3}}}{3} = -\frac{1}{3}$
	Obtains correct $\frac{dy}{dx}$	AO1.1b	A1	
	Substitutes $t = -2$ (or $x = -6$) into 'their' equation for $\frac{dy}{dx}$	AO1.1a	M1	
	Obtains correct simplified gradient of the curve FT 'their' equation for $\frac{dy}{dx}$	AO1.1b	A1F	
(b)	Eliminates t or makes t the subject in one expression (evidence for this includes one equation with t as the	AO1.1a	M1	$t^3 = (x - 2)$, $t^2 = (y + 1)$ $t^6 = (x - 2)^2$, $t^6 = (y + 1)^3$ $(x - 2)^2, = (y + 1)^3$

subject or two equations with equal powers of t.)			
Finds a correct Cartesian equation in any form	AO1.1b	A1	**ALT** $t^3 = (x - 2)$ $t = (x-2)^{\frac{1}{3}}$ $y = (x-2)^{\frac{2}{3}} - 1$
			Total 6 marks

Q16.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Starts an argument by showing that $f(-2) < 0$ and $f(-1) > 0$ Both attempted and at least one evaluated correctly f must be clearly defined or substitution of values must be explicit.	AO2.1	R1	$f(x) = x^3 - 3x + 1$ $f(-2) = (-2)^3 - 3(-2) + 1 = -1 < 0$ $f(-1) = (-1)^3 - 3(-1) + 1 = 3 > 0$ Change of sign and $f(x)$ is continuous so a root must lie between $x = -2$ and $x = -1$
	Explains reasoning fully to complete the argument Evaluations above need to be of opposite sign **and** 'change of sign' OE seen **and** reference to x-values −2 & −1 **and** reference to continuous function	AO2.4	E1	
(b)	Uses Newton–Raphson, must have $f'(x)$ correct PI by correct substitution	AO1.1a	M1	$x_{n+1} = x_n - \frac{(x_n)^3 - 3x_n + 1}{3(x_n)^2 - 3}$
	Substitutes $x_1 = -2$ into 'their' Newton–Raphson formula (accept 'their' $f(-2)$ from part **(a)**)	AO1.1a	M1	$x_2 = -2 - \frac{-1}{3(-2)^2 - 3}$
	Obtains correct value for x_2 $-\frac{17}{9}$ or $-1\frac{8}{9}$ or −1.89 or	AO1.1b	A1	$= -\frac{17}{9}$

	Marking Instructions	AO	Marks	Typical Solution
	better			
(c)	Explains why the method fails when $x_1 = -1$ This must include a substitution of $x_1 = -1$ and an explanation of what goes wrong eg division by zero | not possible division by zero | method fails	AO2.4	E1	$x_1 = -1$ $x_2 = -2 - \frac{3}{3(-1)^2 - 3} = -2 - \frac{3}{0}$ causes division by zero (expression undefined) **ALT** f '(−1) = 0, function has zero gradient at this point, method will fail
				Total 6 marks

Q17.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses the product rule for either term	AO1.1a	M1	$y = 2x\cos 3x + (3x^2 - 4)\sin 3x$ $\frac{dy}{dx} = 2\cos 3x - 6x\sin 3x$ $+6x\sin 3x + 3(3x^2 - 4)\cos 3x$ $= (9x^2 - 10)\cos 3x$
	Uses the product rule for both terms	AO1.1a	M1	
	Differentiates both terms correctly	AO1.1b	A1	
	Rearranges to correct form CAO	AO1.1b	A1	
(b)	Finds $\frac{d^2y}{dx^2}$ from 'their' first derivative and equates to zero	AO3.1a	M1	$\frac{d^2y}{dx^2} = 18x\cos 3x - 3(9x^2 - 10)\sin 3x$
	Applies product rule correctly on 'their' $\frac{dy}{dx}$ FT only applies if both M1 marks awarded in part **(a)**	AO1.1b	A1F	point of inflection ⇒ $\frac{d^2y}{dx^2} = 0$ ⇒ $18x\cos 3x - 3(9x^2 - 10)\sin 3x = 0$ $\Rightarrow \frac{\cos 3x}{\sin 3x} = \frac{3(9x^2 - 10)}{18x}$
	Arrives at a result using 'their' second derivative through correct algebraic manipulation that is correct for 'their' second derivative FT only applies if both	AO1.1b	A1F	$\Rightarrow \cot 3x = \frac{9x^2 - 10}{6x}$ **(AG)**

previous marks in **(b)** have been awarded.			
Constructs a clearly explained rigorous mathematical argument, to show the required result This must include a concluding statement or an explanation of reasoning at the start. **AG**	AO2.1	R1	
			Total 8 marks

Q18.

Marking Instructions	AO	Marks	Typical Solution
Selects a method of integration, which could lead to a correct solution. Evidence of integration by parts OR an attempt at integration by inspection.	AO3.1a	M1	$u = \ln 2x, \quad \frac{dv}{dx} = x^3$ $\frac{du}{dx} = \frac{1}{x}; \quad v = \frac{x^4}{4}$ $\left[\frac{x4}{4}\ln(2x)\right]_1^2 - \int_1^2 \frac{x^3}{4}dx$
Applies integration by parts formula correctly OR correctly differentiates an expression of the form $Ax^4 \ln 2x$	AO1.1b	A1	$\left[\frac{x4}{4}\ln(2x) - \frac{x^4}{16}\right]_1^2$ $= \left(\frac{2^4}{4}\ln(4) - \frac{2^4}{16}\right) - \left(\frac{1}{4}\ln(2) - \frac{1}{16}\right)$
Obtains correct integral, condone missing limits.	AO1.1b	A1	
Substitutes correct limits into 'their' integral	AO1.1a	M1	$\frac{31}{4}\ln 2 - \frac{15}{16}$
Obtains correct p and q FT use of incorrect integral provided both M1 marks have been awarded	AO1.1b	A1F	so $p = \frac{31}{4} \quad q = -\frac{15}{16}$ **ALT** $\frac{d}{dx}(x^4 \ln 2x) = 4x^3 \ln 2x + x^4 . \frac{1}{x}$ $\therefore \int_1^2 x^3 \ln 2x dx = \left[\frac{1}{4}(x^4 \ln 2x - \frac{x^4}{4})\right]_1^2$ $= \left(\frac{2^4}{4}\ln(4) - \frac{2^4}{16}\right) - \left(\frac{1}{4}\ln(2) - \frac{1}{16}\right)$

			$\frac{31}{4}\ln 2-\frac{15}{16}$ $p=\frac{31}{4}\quad q=-\frac{15}{16}$
			Total 5 marks

Q19.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds 2nd derivative and sets up an inequality	AO3.1a	M1	$\frac{dy}{dx}=-2xe^{-x^2}$ $\frac{d^2y}{dx^2}=-2e^{-x^2}+4x^2e^{-x^2}$ $-2e^{-x^2}+4x^2e^{-x^2}<0$ $4x^2-2<0$ $-\frac{\sqrt{2}}{2}<x<\frac{\sqrt{2}}{2}$
	Obtains correct first derivative	AO1.1b	A1	
	Obtains second derivative correct from 'their' first derivative	AO1.1b	A1F	
	Deduces correct final inequality (could use set notation)	AO2.2a	A1	
(b)	Uses trapezium rule	AO1.1a	M1	$\int_{0.1}^{0.5}e^{-x^2}dx\approx\frac{0.1}{2}(e^{-0.01}+e^{-0.25}$ $+2(e^{-0.04}+e^{-0.09}+e^{-0.16}))$ ≈ 0.3611
	Trapezium rule entries all correct	AO1.1b	A1	
	Finds correct value	AO1.1b	A1	
(c)	References area being completely within concave section So...	AO2.4	E1	$[0.1, 0.5]\subset\left[-\frac{\sqrt{2}}{2},\frac{\sqrt{2}}{2}\right]$ ∴ area is completely within concave section Hence trapezia lie below curve and give an under-estimate for the area
	Trapezia all fall completely underneath the curve therefore underestimate (only award this mark if previous E1 has been awarded)	AO2.4	E1	
(d)	Uses suitable rectangle to obtain over-estimate	AO3.1a	B1	Using a rectangle with the left hand edge the same height as the curve will produce an over-estimate Area of rectangle = $0.4\times e^{-0.01^2}=0.396...$ ∴ $0.36<A<0.40$
	Explains that this rectangle lies above the curve	AO2.4	E1	
	Constructs rigorous mathematical argument about accuracy, which	AO2.1	R1	

leads to required result Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.			So A = 0.4 to 1 dp
			Total 12 marks

Q20.

(a) $\frac{8}{x^2} = 8x^{-2}$

PI by its derivative as $16x^{-3}$ or $-16x^{-3}$

B1

$\frac{dy}{dx} = 2 + 16x^{-3}$

Differentiating either 6 + $2x$ correctly or differentiating $-8/x^2$ correctly.

M1

$2 + 16x^{-3}$ OE

A1

3

(b) At $P(2, 8)$, $\frac{dy}{dx} = 2 + 16 \times 2^{-3}$ (= 4)

Attempt to find $\frac{dy}{dx}$ when $x = 2$

M1

Gradient of normal at $P = -\frac{1}{4}$

$m \times m' = -1$ used

m1

Eqn. of normal at P:

$y - 8 = -\frac{1}{4}(x - 2) \Rightarrow x + 4y = 34$

CSO AG

A1

3

(c) (i) At St. Pt $\frac{dy}{dx} = 0$, $2 + 16x^{-3} = 0$

Equating c 's $\frac{dy}{dx}$ *to 0*

Accept ' $\frac{dy}{dx}$ *= 0 so x = −2' stated with no errors seen*

M1

($16x^{-3} = -2$) $x = -2$

$x = -2$

A1

When $x = -2$, $y = 6 - 4 - 2 = 0$;
$M(-2, 0)$ lies on x-axis

Need statement and correct coords.

A1

3

(ii) Tangent at M has equation $y = 0$

$y = 0$ *OE*

B1

1

(d) Intersects normal at P when $x + 0 = 34$

PI Solving c's eqn. of tangent with ans (b) as far as correctly eliminating one variable.

M1

T (34, 0)

Accept x = 34, y = 0

A1

2

[12]

Q21.

(a) $\left(\frac{dy}{dx} = \right) 3e^{3x} + \frac{1}{x}$

B1 for one term correct

B1

B1 all correct

B1

2

(b) (i) $\left(\frac{du}{dx}=\right)\frac{\pm\cos x(1+\cos x)\pm\sin x(\sin x)}{(1+\cos x)^2}$

clear attempt at quotient / product rule
condone poor use of brackets

M1

$\frac{\cos x(1+\cos x)-\sin x(-\sin x)}{(1+\cos x)^2}$

any correct form seen

A1

$= \frac{\cos x+\cos^2 x+\sin^2 x}{(1+\cos x)^2} = \frac{\cos x+1}{(1+\cos x)^2} = \frac{1}{1+\cos x}$

AG be convinced
correct use of brackets and correct notation used throughout (eg A0 if cos x^2 etc seen)

A1cso

3

(ii) $\left(\frac{dy}{dx}=\right)\frac{1+\cos x}{\sin x}\times\frac{1}{1+\cos x}$ OE

correct use of chain rule

M1

$= \frac{1}{\sin x}$

$= \operatorname{cosec} x$

AG, must see $= \frac{1}{\sin x}$ *and no errors seen;*
condone incorrect use of brackets only if penalised in part (b)(i)

A1

2

[7]

Q22.

(a) $\frac{dy}{dx} = 5x^4 - 4x$

one of these terms correct

M1

all correct (no + c etc)

A1

$(= 5(-1)^4 - 4(-1))$ $= 9$

A1

Tangent has equation y = *'their 9'* $x + c$
and 6 = *'their 9'* $(-1) + c \Rightarrow$ =...

tangent using 'their□ gradient, and attempt to find c using $x = -1$ and $y = 6$

m1

$\Rightarrow y = 9x + 15$

equation must be seen in this form

A1

5

(b) (i) When $x = 2$, $y = 2^5 - 2 \times 2^2 + 9 = 32 - 8 + 9 = 33$

*be convinced that they are using **curve** equation*

$(k =)$ 33

***NMS** k = 33 scores B0*

B1

1

(ii) When $x = 2$, $y = 9 \times 2 + 15 = 33$ so lies on tangent

*be convinced that they are using **tangent** equation **and** have statement*

B1

1

(c) (i) $\frac{x^6}{6} - \frac{2x^3}{3} + 9x$

one of these terms correct

M1

another term correct

A1

all correct (may have +c)

A1

$$\left[\frac{2^6}{6} - \frac{2 \times 2^3}{3} + 9 \times 2\right] - \left[\frac{(-1)^6}{6} - \frac{2 \times (-1)^3}{3} + 9 \times (-1)\right]$$

$$\left[\frac{64}{6}-\frac{16}{3}+18\right]-\left[\frac{1}{6}+\frac{2}{3}-9\right]$$

= 31.5

F(2) − F(−1) unsimplified FT "their terms" from integration = $\frac{70}{3}-\left(-\frac{49}{6}\right)$

m1

$\left(or\ \frac{189}{6}\ etc\right)$

condone single fraction

A1

5

(ii) Area of trapezium = $\frac{1}{2}$ × 3 × (6 + 'their' k)

= 58.5 when k = 33

B1√

Shaded area = **Trapezium** − 'their' (c)(i) value

M1

= 27

OE eg $\frac{162}{6}$

A1

3

[15]

Q23.

(a) $\sqrt{x} = x^{0.5}$

$\sqrt{x} = x^{0.5}$ *or* $\sqrt{x} = x^{\frac{1}{2}}$ *seen or used*

B1

$$\frac{12+x^2\sqrt{x}}{x} = \frac{12+x^{2.5}}{x}$$

= $12x^{-1} + x^{1.5}$

$12x^{-1}$ or p = −1

B1

$x^{1.5}$ *or* $q = \frac{3}{2}$ *(=1.5)*

B1

3

(b) (i) $\frac{dy}{dx} = -12x^{-2}$

Ft on c's p only if c's p is a negative integer

B1F

$+ 1.5x^{0.5}$

Ft on c's q only if c's q is a pos non-integer

B1F

2

(ii) When $x = 4$, $y = 11$

B1

When $x = 4$, $\frac{dy}{dx} = \frac{-12}{16} + 3 = \frac{9}{4}$

Attempt to find $\frac{dy}{dx}$ *when x = 4 PI*

M1

Gradient of normal = $-\frac{4}{9}$

$m \times m' = -1$ *used*

m1

Eqn of normal: $y - 11 = -\frac{4}{9}(x - 4)$

ACF eg $4x + 9y = 115$

A1

4

(iii) At St Pt $\frac{dy}{dx} = -12x^{-2} + 1.5x^{0.5} = 0$

Equating c's $\frac{dy}{dx}$ *to zero*

M1

$\Rightarrow x^2 x^{0.5} = 8$, $\Rightarrow x^{\frac{5}{2}} = 8 \Rightarrow x = 8^{\frac{2}{5}}$

A correct eqn in the form $x^n = c$ or $x = c^{\frac{1}{n}}$ correctly obtained.

A1

$\Rightarrow x = (2^3)^{\frac{2}{5}} \Rightarrow x = 2^{\frac{6}{5}}$

CSO $x = 2^{\frac{6}{5}}$. All working must be correct and in an exact form.
If '$x = 0$' also appears then A0 CSO

A1

3

[12]

Q24.

(a) ($y = x^4 \tan 2x$)

$\left(\frac{dy}{dx} =\right)$ $4x^3 \tan 2x + x^4 2\sec^2 2x$

$4x^3 \tan 2x + Ax^4 \sec^2 kx$
OE where A is a non-zero constant.

M1

A1 for $k = 2$
may have $(\sec 2x)^2$
or $\frac{1}{\cos^2 2x}$

A1

A1 all correct
ISW if attempt to simplify is incorrect.

A1

3

(b) $\left(\frac{dy}{dx} =\right)\frac{\pm 2x(x-1) \pm 1(x^2)}{(x-1)^2}$

Use of the quotient rule

M1

$\frac{2x(x-1) - 1(x^2)}{(x-1)^2}$

A1

$\left(= \frac{x^2 - 2x}{(x-1)^2}\right)$

Simplification not required

$\left(\frac{dy}{dx} = \right)\frac{3}{4}$ or 0.75 OE

Obtained from correct $\frac{dy}{dx}$

A1

3

[6]

Q25.

(a) (i) $\left(\frac{dy}{dx} = \right)$ $5x^4 - 6x + 1$

one term correct

M1

another term correct

A1

all correct (no + c etc)

A1

(ii) $\left(\frac{d^2y}{dx^2} = \right)$ $20x^3 - 6$

FT 'their' $\frac{dy}{dx}$

B1√

(b) $x = -1 \Rightarrow \frac{dy}{dx} = 5(-)^4 - 6(-1) + 1$ (= 12)

must sub x= −1 into 'their' $\frac{dy}{dx}$

M1

$\Rightarrow y = 12(x + 1)$

any correct form with (x − −1) simplified
condone y = 12x + c, c=12

A1cso

(c) $x = 1 \Rightarrow \frac{dy}{dx} = 5 - 6 + 1$

sub x = 1 into their $\frac{dy}{dx}$

$\frac{dy}{dx} = 0 \Rightarrow$ stationary point

shown = 0 plus correct statement

A1cso

when $x = 1$, $\frac{d^2y}{dx^2} = 14 \Rightarrow$ (B is a) minimum (point)

or $\frac{d^2y}{dx^2} = 20 - 6 > 0 \Rightarrow$ *(B is a) minimum (point)*

must have correct $\frac{dy}{dx}$ *and* $\frac{d^2y}{dx^2}$ *for E1*

E1

(d) (i) $\frac{x^6}{6} - \frac{3x^3}{3} + \frac{x^2}{2} + 5x$

one term correct

M1

another term correct

A1

all correct (may have + c)

A1

$$\left[\frac{1}{6} - 1 + \frac{1}{2} + 5\right] - \left[\frac{1}{6} + 1 + \frac{1}{2} - 5\right]$$

'their' F(1) − F(−1) with powers of 1 and −1 evaluated correctly

M1

= 8

A1cso

5

(ii) 'their answer to part (i)' − 2

M1

$\Rightarrow$ Area = 6

A1cso

2

[16]

Q26.

(a) $\frac{dy}{dx} = 12 - 5x^{\frac{2}{3}}$

$kx^{\frac{2}{3}}$ term.

M1

ACF

A1

2

(b) (i) When $x = 0$, $\frac{dy}{dx} = 12$

Ft on c's y' evaluated correctly at $x = 0$

B1F

Eqn of tangent at O is $y = 12x$

OE Ft on c's value for $y'(0)$ provided $y'(0) > 0$.

B1F

2

(ii) When $x = 8$, $\frac{dy}{dx} = 12 - 5 \times (8)^{\frac{2}{3}}$

Attempt to find $\frac{dy}{dx}$ when $x = 8$

M1

Equation of tangent at (8, 0) is $y - 0 = y'(8)[x - 8]$

$y = y'(8)[x - 8]$ OE

M1

$y = -8(x - 8) \Rightarrow y + 8x = 64$

CSO AG

A1

3

(c) $\int \left(12x - 3x^{\frac{5}{3}}\right) dx = \frac{12x^2}{2} - \frac{3x^{\frac{8}{3}}}{\frac{8}{3}} \; (+c)$

$kx^{\frac{5}{3}+1}$ term after integrating, condone k left unsimplified for this M mark.

$= 6x^2 - \frac{9}{8}x^{\frac{8}{3}} \; (+c)$

For $6x^2$ OE eg $(12x^2 / 2)$

B1

For $-\frac{9}{8}x^{\frac{8}{3}}$ *OE*

A1

3

(d) Area bounded by curve and x-axis

$= \int_0^8 \left(12x - 3x^{\frac{5}{3}}\right) \mathrm{d}x = 6 \times 8^2 - \frac{9}{8} \times (8)^{\frac{8}{3}}$

± F(8) {- F(0)} PI following integration

M1

= 384 − 288 = 96

PI by correct final answer if evaluation not seen here

A1

At P, $12x + 8x = 64$

Solving $y + 8x = 64$ and c's $y = kx$, $k > 0$, down to an eqn in one variable…

$[y + 2y / 3 = 64]$

M1

$(x_p = 3.2)$ $y_p = 38.4$

For y_p38.4 OE [If using integration to find area of triangle, award A1 if both 'x_p = 3.2' and correct integration of correct eqns of the 2 lines]

A1

Area of triangle OPA = $\frac{1}{2} \times 8 \times y_p$

OE Need perpendicular ht to be linked to $y_p > 0$.

M1

Area of shaded region = Area Δ OPA − $\int_0^8 \left(12x - 3x^{\frac{5}{3}}\right) \mathrm{d}x$

M0 if evaluated to a value < 0

M1

$= 153.6 - 96 = 57.6$

OE eg 288 / 5

A1

7

[17]

Q27.

(a) $\left(\frac{dy}{dx} = \right)$ $12x^2 - 6$

do not ISW

B1

1

(b) $\int_2^3 \frac{2x^2 - 1}{4x^3 - 6x + 1}\,dx = \left[\frac{1}{6}\ln\left(4x^3 - 6x + 1\right)\right]_{(2)}^{(3)}$

k ln $(4x^3 - 6x + 1)$, k is a constant

M1

$k = \frac{1}{6}$

A1

$= \frac{1}{6}$ ln $(4 \times 3^3 - 6 \times 3 + 1)$

$- \frac{1}{6}$ ln $(4 \times 2^3 - 6 \times 2 + 1)$

correct substitution in F(3) − F(2).
condone poor use or lack of brackets.

m1

$= \frac{1}{6}$ ln 91 − $\frac{1}{6}$ ln 21

k ln 91 − k ln 21
only follow through on their k

A1F

$= \frac{1}{6}\ln\frac{91}{21}$ or $\left(= \frac{1}{6}\ln\frac{13}{3}\right)$

A1

or if using the substitution

$u = 4x^3 - 6x + 1$

$\int = k\int\frac{du}{u}$ *M1*

$= \frac{1}{6}\ ln\ u$ A1

then, either change limits to 21 and 91 m1 then A1F A1 as scheme or changing back to 'x', then m1 A1F A1 as scheme

5

[6]

Q28.

(a) $\left(\frac{dx}{d\theta}=\right)\frac{(\sin\theta\times 0)-1\times\cos\theta}{\sin^2\theta}$

quotient rule $\frac{\pm\sin\theta\times k\pm 1\times\cos\theta}{\sin^2\theta}$ *where* k *= 0 or 1*

M1

must see the '0' either in the quotient or in eg $\frac{du}{d\theta}$ *= 0 etc*

A1

$= -\frac{\cos\theta}{\sin^2\theta}$ or $= -\frac{\cos\theta}{\sin\theta\sin\theta}$

or equivalent

$= -\ \text{cosec}\theta\ \cot\theta$

CSO, AG must see one of the previous expressions

A1

(b) $x = \text{cosec}\theta$

$\frac{dx}{d\theta} = -\text{cosec}\theta\ \cot\theta$

OE, eg dx = − cosecθ cotθ dθ

B1

Replacing $\sqrt{(\text{cosec}^2\theta-1)}$ by $\sqrt{\cot^2\theta}$, or better

at any stage of solution

B1

$\int=\int\frac{-\text{cosec}\theta\cot\theta}{\text{cosec}^2\theta\sqrt{(\text{cosec}^2\theta-1)}}\ d\theta$

all in terms of θ, and including their attempt at dx, but condone omission of dθ

M1

fully correct and must include dθ (at some stage in solution)

A1

$\int \frac{-\operatorname{cosec}\theta \cot\theta}{\operatorname{cosec}^2\theta \cot\theta}\ (\mathrm{d}\theta) = \int \frac{-1}{\operatorname{cosec}\theta}\ (\mathrm{d}\theta)$

OE eg $\int -\sin\theta\ (d\theta)$

A1

$= \cos\theta$

A1

$x = 2$, $\theta = 0.524$ AWRT

$x = \sqrt{2}$, $\theta = 0.785$ AWRT

correct change of limits or $(\pm)\cos\theta = (\pm)\left[\sqrt{\left(1-\frac{1}{x^2}\right)}\right]_{\sqrt{2}}^{2}$ *OE*

B1

0.8660 − 0.7071

c's F(0.52) − F(0.79)

substitution into ± cosθ only or $\left(\frac{\sqrt{3}}{2}-\frac{1}{\sqrt{2}}\right)$

m1

= 0.159

A1

9

[12]

Q29.

(a) (i) (When $x = 2$) $\frac{dy}{dx} = 12 - 1 - 11 = 0$

AG Must see intermediate evaluations

B1

1

(ii) $\frac{4}{x^2} = 4x^{-2}$ {so $\frac{dy}{dx} = 3x^2 - 4x^{-2} - 11$}

$\frac{4}{x^2} = 4x^{-2}$*, seen in (a)(ii) or earlier. PI by* $\pm 8x^{-3}$ *term in answer*

B1

$\frac{d^2y}{dx^2} = 6x + 8x^{-3}$

Correct powers of x correctly obtained from differentiating the first two terms

M1

$6x + 8x^{-3}$ ACF

A1

When $x = 2$, $\frac{d^2y}{dx^2} = 12 + 8/8 = 13$

A1

4

(iii) Since $\frac{d^2y}{dx^2} > 0$, P is a minimum point.

Ft on c's value of $y''(2)$ in (a)(ii) but must see reference to sign of $y''(2)$ either explicitly or as inequality, as well as the correct ft conclusion

E1F

1

(b) $\int \left(3x^2 - \frac{4}{x^2} - 11\right) dx = x^3 + 4x^{-1} - 11x\ (+c)$

Attempt to integrate $\frac{dy}{dx}$ with at least two of the three terms integrated correctly

M1

$(y =)\ x^3 + 4x^{-1} - 11x(+c)$

For $x^3 + 4x^{-1} - 11x$ OE even unsimplified.

1

When $x = 2$, $y = 1 \Rightarrow 1 = 8 + 2 - 22 + c$

Substituting $x = 2$, $y = 1$ into $y = F(x) +$ 'c' in attempt to find constant of integration, where $F(x)$ follows attempted integration of expression for $\frac{dy}{dx}$

$y = x^3 + 4x^{-1} - 11x + 13$

ACF

A1

4

[10]

Q30.

(a) $\left(\frac{dy}{dx} =\right)$ $x^3 \times \frac{1}{x} + 3x^2 \ln x$

$px^3 \times \frac{1}{x} + qx^2 \ln x$

where p and q are integers

M1

$p = 1, q = 3$

A1

2

(b) (i) $\left(\frac{dy}{dx} =\right)$ $e^2 + 3e^2 \ln e \quad (= 4e^2)$

Substituting e for x in their $\frac{dy}{dx}$, but must have scored M1 in (a)

M1

$y = e^3 \ln e \ (= e^3)$

B1

$y - e^3 = 4e^2 (x - e)$

*OE **but** must have evaluated ln e (twice) for this mark (must be in exact form, but condone numerical evaluation after correct equation)*

A1

3

(ii) $- e^3 = 4e^2 (x - e)$ or $4e^2 x = 3e^3$ OE

Correctly substituting $y = 0$ into a correct tangent equation in (b)(i)

M1

$x = \frac{3}{4} e$

CSO;

ignore subsequent decimal evaluation

A1

2

[7]

Q31.

(a) $\frac{dy}{dx} = 18 + 6x - 12x^2$

one of these terms correct

M1

another term correct

A1

all correct (no + c etc)
(penalise + c once only in question)

A1

3

(b) $18 + 6x - 12x^2 = 0$

putting their $\frac{dy}{dx} = 0$,

PI by attempt to solve or factorise

M1

$6(3 - 2x)(x + 1) \quad (= 0)$

attempt at factors of ***their quadratic***
or use of quadratic equation formula

m1

$x = -1, \quad x = \frac{3}{2}$ OE

must see both values unless $x = -1$
is verified separately

A1

3

If M1 not scored, award SC B1 for verifying

that $x = -1$ leads to $\frac{dy}{dx}$ *= 0 and a further*

SC B2 for finding $x = \frac{3}{2}$ *as other value*

(c) (i) $\frac{d^2y}{dx^2} = 6 - 24x$

FT their $\frac{dy}{dx}$ *but* $\frac{d^2y}{dx^2}$ *must be correct if*

3 marks earned in part (a)

B1 √

When $x = -1$, $\frac{d^2y}{dx^2} = 6 - (24 \times -1)$

Sub $x = -1$ *into their* $\frac{d^2y}{dx^2}$

M1

$\frac{d^2y}{dx^2} = 30$

A1cso

3

(ii) Minimum point

must have a value in (c)(i)

FT "maximum" if their value of $\frac{d^2y}{dx^2} < 0$

E1 √

1

[10]

Q32.

(a) (i) $\frac{dy}{dx} = -1 - 4x^3$

one of these terms correct

M1

all correct (no + c*)*

A1

(When $x = 1$, grad =) -5

(Check that $\frac{dy}{dx}$ *is actually correct!)*

A1cso

3

(ii) $y - 12$ = 'their grad' $(x - 1)$

any form of equation through (1, 12) and attempt at c *if using* $y = mx + c$

M1

$y = -5x + 17$ (or $y = 17 - 5x$)

FT their gradient

Condone $y = -5x + c$, $c = 17$ etc

A1 √

2

(b) (i) $14x - \frac{x^2}{2} - \frac{x^5}{5}$

one of these terms correct

M1

another term correct

A1

all correct (may have + c)

A1

$[\;]_{-2}^{1} =$

$\left(14 - \frac{1}{2} - \frac{1}{5}\right) - \left(-28 - 2 + \frac{32}{5}\right)$

F(1) and F(–2) attempted

m1

= 36.9 OE

Condone recovery to this value

A1

5

(ii) Area $\Delta = \frac{1}{2} \times 3 \times 12$

Correct area of triangle unsimplified

M1

= 18

$\Rightarrow$ shaded area = 18.9

A1cso

2

[12]

Q33.

(a) $y = x + 3 + \frac{8}{x^4} = x + 3 + 8x^{-4}$

For $\frac{8}{x^4} = 8x^{-4}$ *PI by correct differentiation of 3rd term*

B1

$\frac{dy}{dx} = 1 - 32x^{-5}$ *or* $1 - \frac{32}{x^5}$

$k\,x^{-5}$ *OE*

M1

For either

A1

3

(b) When $x = 1$, $y = 12$

B1

When $x = 1$, $\frac{dy}{dx} = 1 - 32 = -31$

Attempt to find value of $\frac{dy}{dx}$ *when* $x = 1$

M1

Tangent: $y - 12 = -31(x - 1)$

*Only ft on c's answer to (**a**). Any correct (ft on c's (a)) form.*

A1F

3

(c) $1 - 32x^{-5} = 0$

$1 - 32x^{-5} = 0$ *or c's* $\frac{dy}{dx} = 0$

M1

$\Rightarrow x^5 = 32$

Attempt to form x^n *= const (≠0). PI by next line*

m1

$\Rightarrow x = 2$

CSO

A1

(Coordinates of M) (2, 5.5)

CSO

A1 4

(d) (i) $\int\left(x+3+\frac{8}{x^4}\right)\mathrm{d}x$

$= \frac{x^2}{2}+3x-\frac{8}{3}x^{-3}+c$

Power −3 correctly obtained

M1

$-\frac{8}{3}x^{-3}$

A1

$\frac{x^2}{2}+3x+c$

B1 3

(ii)

$= \left(2+6-\frac{1}{3}\right)-\left(\frac{1}{2}+3-\frac{8}{3}\right)$

Attempting to calculate F(2) − F(1) where F(x) is c's answer to part (d)(i) provided F is not just the c's integrand (x + 3 + 8/x^4)

M1

$\frac{9}{2}+\frac{7}{3}=\frac{41}{6}$

OE Accept 6.83 or better provided d(i) used

A1 2

(e) $k = -5.5$

Ft on $-y_M$ from part (c).

B1F 1

[16]

Q34.

(a) $\frac{dy}{dx} = k(x^3 - 1)^5$

Where k is an integer or function of x

M1

$= 6 \times 3x^2(x^3 - 1)^5$ (ISW)

But note

$\frac{dy}{dx} = k(x^3 - 1)^5 + px^2$ *M0*

Or

($u = x^3 - 1$) *($y = u^6$)*

$\frac{dy}{du} = 6u^5$ *and* $\frac{du}{dx} = 3x^2$ *M1*

$= 6(x^3 - 1)^5 \times 3x^2$ *A1*

Note

$\frac{dy}{dx} = 6 \times 3x^2(x^3 - 1)^5 + c$ *scores* *M1 A0*

(penalise + c in differential once only in paper)

A1

2

(b) (i) $\frac{dy}{dx} = \pm x \times \frac{1}{x} \pm \ln x$

*Product rule attempted **and** differential of ln x*

M1

$= 1 + \ln x$ (ISW)

A1

2

(ii) (x = e) y = e PI

Must have replaced ln e by 1
Condone y = 2.72 (AWRT)

B1

$\frac{dy}{dx} = 1 + \ln e (= 2)$

Correct substitution into their $\frac{dy}{dx}$
But must have scored M1 in (b)(i)

M1

y – e = 2(x – e) or y = 2x – e OE, ISW

Must have replaced ln e by 1

A1

3

[7]

Q35.

(a) *For either 6 or 6x^0*

B1

$$\frac{dy}{dx} = 6 - 3x^{\frac{1}{2}}$$

$Ax^{\frac{3}{2}-1}$, *$A \neq 0$ OE*

M1

$6 - 3x^{\frac{1}{2}}$ *or* $6 - 3\sqrt{x}$ *with no '+c'*

[If unsimplified here, A1 can be awarded retrospectively if correct simplified expression is seen explicitly in (b)(i).]

A1

3

(b) (i) $6 - 3x^{\frac{1}{2}} = 0$

Equating c's $\frac{dy}{dx}$ *to 0 PI by correct ft rearrangement of c's dy/dx = 0*

M1

$x^{\frac{1}{2}} = 2 \Rightarrow x = 2^2$

$x^{\frac{1}{2}} = k$ *(k > 0), to $x = k^2$.*

PI by correct value of x if no error seen

m1

M(4, 8)

SC If M0 award B1 for (4, 8)

A1

3

(ii) Eqn of normal at M is $x = 4$

Ft on $x = c$'s x_M

B1F

1

(c) (i) When $x = \frac{9}{4}, \frac{dy}{dx} = 6 - 3 \times \frac{3}{2} = \frac{3}{2}$

Attempt to find $\frac{dy}{dx}$ when $x = \frac{9}{4}$

M1

Gradient of normal at P = $-\frac{2}{3}$

$m \times m' = -1$ used

m1

Eqn of normal: $y - \frac{27}{4} = -\frac{2}{3}\left(x - \frac{9}{4}\right)$

ACF eg $y = -\frac{2}{3}x + \frac{33}{4}$

A1

$12y - 81 = -8x + 18 \Rightarrow 8x + 12y = 99$

Coeffs and constant must now be positive integers, but accept different order

eg $12y + 8x = 99$

A1

4

(ii) $8(4) + 12y = 99$

Solving c's answer (b)(ii), (must be in form x = positive const), with c's answer (c)(i).
PI by correct earlier work and correct coordinates for R.

M1

$R\left(4, \frac{67}{12}\right)$

Accept 5.58 or better as equivalent to $\frac{67}{12}$

A1

2

[13]